Group Theory.

Defn: A group is a nonempty set $\langle G, \circ \rangle$ such that the following properties hold.

(1) For $\forall a, b \in G$, $a \circ b \in G$ (Closure)

(2) For $\forall a, b, c \in G$

$$a \circ (b \circ c) = (a \circ b) \circ c$$ (Associative).

(3) There must exist an element $e \in G$ such that $e \circ a = a \circ e = a$, $\forall a \in G$.

(Identity: "e" as the identity element).

(4) For each $a \in G$, there is an element $b \in G$ such that $a \circ b = b \circ a = e$ (Existence of inverse).

'b' is the inverse of 'a'

↳ we replace 'b' by a^{-1}

Def^{n}: A group $\langle G, \circ \rangle$ is commutative / abelian

if $a \circ b = b \circ a$.

↓

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E.g.

$\langle \mathbb{Z}, + \rangle \longrightarrow$ group $\longrightarrow$ abelian.

$\langle \mathbb{R}, + \rangle \longrightarrow$ group $\longrightarrow$ abelian.

$\langle \mathbb{Z}, * \rangle \longrightarrow$ X 0 does not have an inverse.

$\langle \mathbb{Z}^*, * \rangle \longrightarrow$

↓

non-zero elements of $\mathbb{Z}$

$\langle \mathbb{Z}^+, * \rangle \longrightarrow$ X

$\langle \mathbb{R}^*, * \rangle \longrightarrow$ ✓ group.

E.g.2. $\mathbb{Z}_n$ | modulo n set. $\begin{vmatrix} n > 1 \\ n \in \mathbb{Z}^+ \end{vmatrix}$

$\langle \mathbb{Z}_6, + \rangle$ ✓

+ mod 6	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$(5+3) \bmod 6$.

$(5+5) \bmod 6$.

↳ Closure ✓

↳ 0

↳ inverse of 2 = 4.

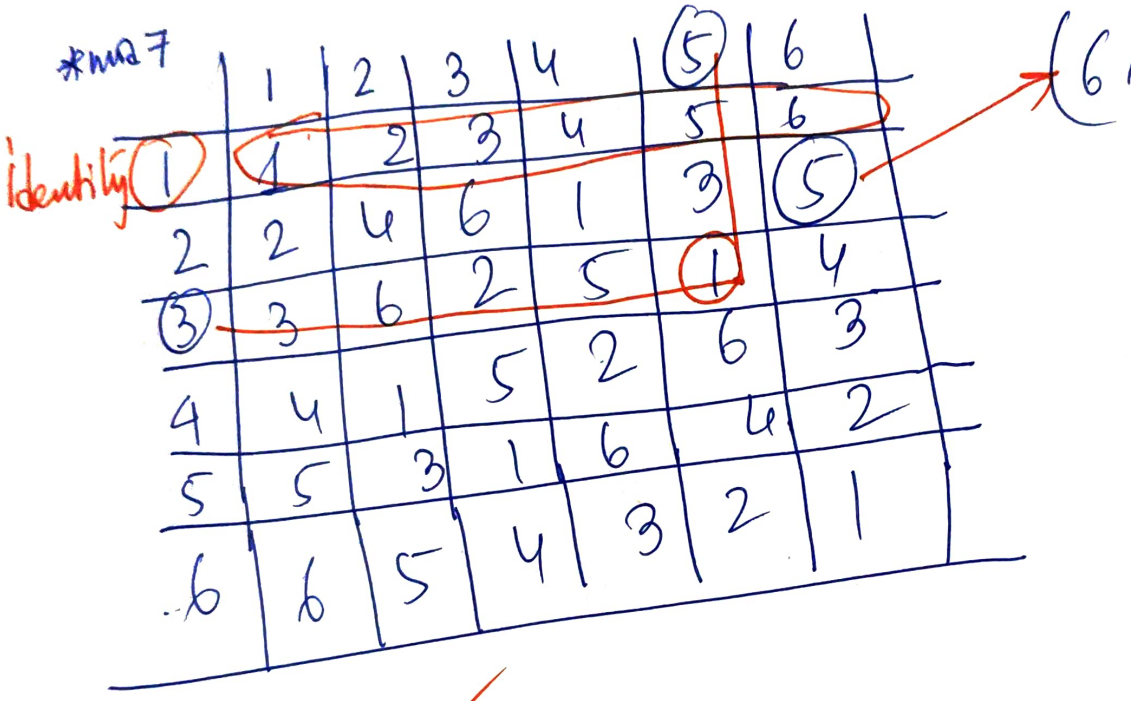
inverse of 4 = 5.

↳ first find $7 \bmod 6 = 1$.

Then find inverse of 1.

E.g. $\mathbb{Z}_p^*$ | p is a prime no.

$\langle \mathbb{Z}_p^*, * \rangle$



Closure → ✓

Associative → ✓

Identity → ✓

Inverse →

Defⁿ:

The number of elements in a group G is called the order of G.

Denoted by: $|G|$

If the set G is infinite, then the order is infinite.

E.g.

$n \in \mathbb{Z}^+$, $|\langle \mathbb{Z}_n, + \rangle| = n$.

$|\langle \mathbb{Z}_p^*, * \rangle| = p-1$ for any prime p.

Theorem:

The identity element of a group $\langle G, \circ \rangle$ is unique.

Contradiction. Assume there are two identity elements e_1 & e_2

$e_1 = e_1 e_2 = e_2$ (by property of identity element)

$\therefore$ Identity element is unique for G.

Theorem:

The inverse of each element in G is unique.

By way of contradiction.

$a \in G$.

a has two inverses: a_1 & a_2.

$aa_1 = e$ [$\because$ a_1 is inverse of a].

$a_2(aa_1) = a_2 e$

$(a_2 a)a_1 = a_2$ [$\because$ a_2 is inverse of a].

$ea_1 = a_2$

$a_1 = a_2$

Theorem:

$$(a^{-1})^{-1} = a$$

$a \circ a^{-1} = a^{-1} \circ a = e$.

$\therefore (a^{-1})^{-1} = a$.

Theorem:

Left & right cancellation.

for $a, b, c \in G$ & $ab = ac$

$\Rightarrow b = c$. [left cancellation].

$$ab = ac$$
$$a^{-1}(ab) = a^{-1}(ac)$$
$$(a^{-1}a)b = (a^{-1}a)c$$
$$eb = ec$$
$$b = c.$$

for $a, b, c \in G$ & $ba = ca$

$b = c$ [right cancellation].

Def^{n}.

Subgroups.

$G = \langle \mathbb{Z}_6, + \rangle$.

$H = \{0, 2, 4\}$.

$+_{mod\,6}$	0	2	4
0	0	2	4
2	2	4	0
4	4	0	2

Is H a group?

↓

yes.

$H \subseteq G$.

Thus H is a subgroup of G.

Let G be a group and $\emptyset \neq H \subseteq G$. If H is a group in itself under the same operation as G, then H is subgroup.

$\{e\}$ is always a subgroup of G. } trivial subgroups of G.

G is " a " of G. }

Theorem: If H is a non empty subset of G

then H is a subgroup of G iff

(a) for $\forall a, b$, $ab \in H$.

(b) for $\forall a$, $a \in H \Rightarrow a^{-1} \in H$.

$\Rightarrow$ H is a subgroup.

} Automatically true.

$\Leftarrow$ $\emptyset \neq H \subseteq G$

~~Associativity~~ ?

Closure: by (a).

Associativity: ? Inherited from G.

Identity: (a) & (b)

$a \in H$, $a^{-1} \in H$ $\rightarrow$ (b)

$ab \in H$, $\forall a, b \in H$

$\downarrow$

$a a^{-1} \in H$

$\Rightarrow e \in H$.

Inverse: by (b)

proved. $\square$.

Theorem: *H is finite.*

If G is a group & $\emptyset \neq H \subseteq G$, with H finite, then H is a subgroup of G iff H is closed under the binary operation of G.

$\Rightarrow$ H is a subgroup of G.
↳ H is closed under the same binary operation of G.

$\Leftarrow$ H is finite & non-empty
(1) H is associative by inheritence.

$a \in H$.

Since H is closed we can say that $a, a^2, a^3, a^4 \ldots$ are in H.
H is finite: Not all of the above are unique.
Thus $a^m = a^n$, $m > n \geq 1$.

Now we take $(a^n)^{-1}$ from G & operate it on both sides:

$$a^{m-n} = a^0 = e$$

Since $m-n > 0$, $a^{m-n} \in H$

Call, $m-n = k$.

$a^k = e \in H$.

Inverse:

$k=1$. $a = e$ $a^{-1} = e \in H$.

$k \geq 2$. $k-1 > 0$.

$a^k = a \cdot a^{k-1}$

$a^{k-1} \in H$.

a^{k-1} is the inverse of a.

□.

Defⁿ:

If $\langle G, \circ \rangle$, $\langle H, * \rangle$

$f : G \to H$ then f is called a group homomorphism if

$\forall a, b \in G$, $f(a \circ b) = f(a) * f(b)$.

Theorem:

$(G, \circ)$ identity element e_G $\quad$ $(H, *)$ identity element e_H

(a) $f(e_G) = e_H$

(b) $f(a^{-1}) = [f(a)]^{-1}$ $\forall a \in G$.

(c) $f(a^n) = [f(a)]^n$ $\forall a \in G$ & $n \in \mathbb{Z}$

(d) $f(S)$ is a subgroup of H for each subgroup S of G.

$$e_H * f(a_H) = f(e_H)$$

$$= f(e_H \circ e_H)$$

$$= f(e_H) * f(e_H)$$

$$e_H * f(e_H) = f(e_H) * f(e_H)$$

Right cancellation:

$$e_H = f(e_H)$$ proved.

Defn: Permutation Groups.

Given a set A, a permutation of A is a function $\sigma : A \to A$ & is a bijection.

If σ is a permutation of a set $M = \{x_1, x_2, \ldots, x_n\}$

$$\sigma = \begin{pmatrix} x_1 & x_2 & \cdots & x_n \\ \sigma(x_1) & \sigma(x_2) & \cdots & \sigma(x_n) \end{pmatrix}$$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 4 & 3 & 1 \end{pmatrix}$$

$\rightarrow \sigma(1) = 2$
$\sigma(2) = 5$
$\sigma(3) = 4$
$\sigma(4) = 3$
$\sigma(5) = 1.$

$1 \to 2 \to 5 \to 1$

$3 \to 4 \to 3$

$\sigma = (1\,2\,5)(3\,4)$ $\longrightarrow$ cyclic notation.

Defn:

Product of permutations.

Two permutations function σ, π

product $\longrightarrow$ compositions of functions.

$\sigma(\pi(x))$.

$$P = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 1 & 3 & 5 \end{pmatrix} \qquad Q = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \end{pmatrix}.$$

QP (Left $\rightarrow$ Right).

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 1 & 3 & 5 \end{pmatrix}.$$

$1 \rightarrow 5 \rightarrow 5$

$2 \rightarrow 4 \rightarrow 3$

$3 \rightarrow 3 \rightarrow 1$

$4 \rightarrow 2 \rightarrow 4$

$5 \rightarrow 1 \rightarrow 2$

$$\rightarrow \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 1 & 4 & 2 \end{pmatrix}$$

Defn Symmetric group (S_n).

S_n consists of all permutations of the set $\{1, 2, \ldots n\}$.

$$S_3 = \{ \underset{\alpha}{\begin{pmatrix}1&2&3\\1&2&3\end{pmatrix}}, \underset{\beta}{\begin{pmatrix}1&2&3\\2&1&3\end{pmatrix}}, \begin{pmatrix}1&2&3\\1&3&2\end{pmatrix}, \begin{pmatrix}1&2&3\\3&2&1\end{pmatrix}, \begin{pmatrix}1&2&3\\3&1&2\end{pmatrix}, \begin{pmatrix}1&2&3\\2&3&1\end{pmatrix} \}.$$

$\longrightarrow$ group under composition

Identity: $\begin{pmatrix}1&2&3\\1&2&3\end{pmatrix}$

Inverse: $\begin{pmatrix}1&2&3\\3&1&2\end{pmatrix}^{-1} = \begin{pmatrix}1&2&3\\2&3&1\end{pmatrix}$.

In general, any S_n is a group.

Q: Is S_n necessarily abelian? ✗

$b = c$.

$a * b = a * c$.

Binary operations.

$a \times a \xrightarrow{*} a$.

well-defined.

$\frac{1}{2} * a = \frac{1}{2} * b$. ✓

$\frac{2}{4} * a = \frac{2}{4} * b$. ✓

} should remain the same.